

CHEONGJU, Korea, Republic of

WMO: 471310

Lat: 36.6392N

Lon: 127.4406E

Elev: 59

StdP: 100.62

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-10.5	-8.6	-19.3	0.7	-6.1	-17.4	0.8	-5.2	7.3	-1.9	6.3	-1.9	1.4	290	0.377

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	6.9	33.1	24.5	31.9	23.8	30.6	23.1	25.9	30.1	25.3	29.4	24.7	28.8	2.7	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
24.8	19.9	27.9	24.2	19.2	27.5	23.5	18.5	27.3	80.5	30.2	77.9	29.5	75.6	28.8	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
6.0	5.0	4.3	-13.1	34.6	2.5	1.4	-14.9	35.6	-16.3	36.4	-17.7	37.1	-19.5	38.1		
			-14.1	26.5	2.2	1.1	-15.7	27.2	-16.9	27.9	-18.2	28.5	-19.7	29.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	13.4	-1.4	1.2	6.6	13.2	18.9	23.1	26.1	26.5	21.6	15.2	7.8	0.8	(6)
(7)		DBStd	10.26	3.82	3.81	3.76	3.63	2.66	2.05	2.31	2.54	2.74	3.34	4.08	3.96	(7)
(8)		HDD10.0	1107	353	247	117	12	0	0	0	0	4	89	285		(8)
(9)		HDD18.3	2607	611	480	363	157	26	1	0	5	105	316	543		(9)
(10)		CDD10.0	2338	0	1	12	108	274	393	500	513	348	166	24	0	(10)
(11)		CDD18.3	796	0	0	0	4	42	143	242	254	103	8	0	0	(11)
(12)		CDH23.3	6671	0	0	0	43	314	966	2218	2513	590	27	0	0	(12)
(13)		CDH26.7	2079	0	0	0	3	48	236	743	937	112	1	0	0	(13)
(14)	Wind	WSAvg	1.7	1.5	1.7	1.9	2.0	1.9	1.7	1.8	1.7	1.6	1.4	1.4	1.4	(14)
(15)	Precipitation	PrecAvg	1208	30	33	51	81	86	138	275	249	143	46	49	27	(15)
(16)		PrecMax	1676	100	99	171	187	209	362	536	572	348	157	128	69	(16)
(17)		PrecMin	850	1	0	2	14	19	5	128	80	5	3	11	2	(17)
(18)		PrecStd	231	26	26	33	51	46	97	102	129	90	39	30	15	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.6	14.8	20.5	26.7	29.9	31.6	34.9	35.1	30.9	25.8	20.7	12.9	(19)
(20)		MCWB	6.8	10.2	11.3	15.3	17.0	19.7	25.3	24.9	22.8	17.7	14.0	8.3		(20)
(21)		2%	DB	7.5	11.6	17.8	24.5	27.9	30.3	33.0	33.6	29.2	23.9	18.0	10.3	(21)
(22)		MCWB	3.8	6.1	9.7	14.1	16.6	19.7	24.6	24.6	21.3	16.4	12.7	6.3		(22)
(23)		5%	DB	5.7	9.6	15.6	22.5	26.6	29.2	31.7	32.4	27.9	22.5	16.2	8.5	(23)
(24)		MCWB	2.5	4.8	8.4	12.8	16.1	19.4	24.2	24.3	20.6	15.7	11.3	5.0		(24)
(25)		10%	DB	4.3	7.4	13.6	20.4	25.1	28.0	30.4	31.1	26.6	21.0	14.4	6.8	(25)
(26)		MCWB	1.4	3.3	7.5	12.0	15.5	19.3	23.8	24.0	19.9	15.0	10.1	4.0		(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.7	11.1	13.4	18.6	21.1	24.0	26.5	26.8	25.1	20.4	16.1	9.1	(27)
(28)		MCDWB	9.1	13.6	17.0	23.2	24.0	26.7	31.2	31.5	28.6	23.0	18.9	12.1		(28)
(29)		2%	WB	4.6	6.9	11.5	16.2	18.9	22.7	25.7	26.1	23.6	18.5	13.7	6.9	(29)
(30)		MCDWB	6.9	10.2	16.0	22.4	24.3	25.9	30.0	30.5	26.7	21.7	17.0	9.4		(30)
(31)		5%	WB	3.0	5.1	9.5	14.5	17.9	21.9	25.2	25.5	22.5	17.1	12.1	5.5	(31)
(32)		MCDWB	5.0	8.9	14.1	19.8	23.4	25.8	29.4	29.7	25.9	20.5	15.0	8.1		(32)
(33)		10%	WB	1.6	3.7	8.0	13.1	17.0	21.1	24.7	25.0	21.4	15.9	10.6	4.1	(33)
(34)		MCDWB	4.0	7.0	12.8	18.6	22.6	25.4	28.8	29.0	25.0	19.7	13.9	6.8		(34)
(35)	Mean Daily Temperature Range		MDBR	7.8	8.9	10.1	11.3	10.5	8.5	6.6	6.9	8.3	9.9	9.0	7.9	(35)
(36)		5% DB	MCDBR	9.0	11.8	13.5	14.6	13.0	10.7	8.3	8.4	9.5	11.3	11.3	10.0	(36)
(37)		MCWBR	6.2	7.3	7.2	6.2	4.6	3.4	2.7	2.6	3.3	4.8	6.4	6.9	(37)	
(38)		5% WB	MCDBR	8.0	10.4	11.1	11.5	9.5	7.4	6.8	6.9	7.1	8.4	9.1	9.2	(38)
(39)		MCWBR	6.2	7.3	7.0	6.1	4.5	3.3	2.6	2.7	3.3	4.7	6.3	6.9	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.436	0.482	0.567	0.598	0.610	0.678	0.631	0.571	0.496	0.480	0.464	0.432	(40)	
(41)		taud	1.944	1.870	1.715	1.696	1.712	1.636	1.793	1.936	2.077	1.993	1.972	1.975	(41)	
(42)		Ebn at Noon	705	724	701	710	711	662	691	724	758	720	671	677	(42)	
(43)		Edh at Noon	141	172	220	236	236	254	216	183	151	150	137	129	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.49	3.24	4.25	4.92	5.55	5.08	4.15	4.31	4.00	3.56	2.50	2.21	(44)	
(45)		RadStd	0.14	0.26	0.43	0.40	0.49	0.43	0.57	0.62	0.45	0.28	0.29	0.16	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (54 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+83	+45

Nomenclature: See separate page